

Common requirements from MSO users

Upgradability for bandwidths is not usually asked for, although additional digital channels requirement is an upgrade that is often requested.

An interesting, highly-demanded feature is a full-fledged arbitrary function/waveform generator. In this era of mixed-signal designs, it is becoming very important for an MSO to have an integrated signal source. Karl Chen, product marketing manager, Rigol Technologies Inc., says, "Inbuilt arbitrary waveform generation, or Arb, is another significant advancement. Crowded work benches and limited budgets are driving users to demand more functionality in one instrument." For example, he adds, "An education lab can readily build an automatic frequency-response test system. Likewise, a time-synchronous circuit could use the Arb to simulate pulses for time-correlation. The possibilities are limitless."

LeCroy's CAN FD TDM oscilloscopes, and RTE and RTO oscilloscopes from Rohde & Schwarz.

MSOs powered with PAM-4 analysis capability

Many known oscilloscope manufacturers have recently introduced pulse amplitude modulation (PAM-4) analysis capability in their range of MSOs.

Mobile computing applications are requiring ever-increasing Internet speeds as well as improved Internet infrastructure. To enable this, very-high-speed connectivity among server systems is absolutely necessary.

Non-return-to-zero (NRZ) encoding is mostly used for modulation in conventional communication methods. Industry experts feel NRZ will not function in a 56Gbit/s environment. Hence, one solution for this problem is to change the technique from NRZ to PAM. This would allow engineers to significantly expand the amount of data these can transmit across high-speed digital communication links. Many industry standards bodies are actively promoting PAM-4 technology.

PAM-4 signalling helps engineers to address design and measurement challenges swiftly and achieve faster time-to-market. Vendors usually provide an additional software tool that can be combined with particular models of MSOs to activate PAM-4 analysis capability.

Recently, Keysight introduced Keysight N8827A PAM-4 analysis software (for V- and Z-series oscilloscopes) and N8827B PAM-4 analysis software (for S-series oscilloscopes), which engineers can use to perform a complete analysis of electrical PAM-4 signals.

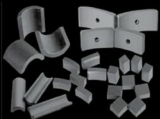
Teledyne LeCroy too released a set of tools for similar analysis and they claim to be the only manufacturer who offer a software that compares jitter, eye and noise measurements on up to four lanes, simultaneously.

Serial-bus triggering trickling down to low-end MSOs

Another trend in MSOs is that serial-bus triggering and decoding is moving down to low-end MSOs. Earlier, only



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